

Legendre's Conjecture in Function Fields: the Classical Range, Full Monodromy, and the Open Range $q \leq d - 2$

Version v7 of *The Geometry of Prime Vacuums*

Ruqing Chen

GUT Geoservice Inc., Montréal, Canada

ruqing@hotmail.com

October 2026

DOI: [10.5281/zenodo.23185896](https://doi.org/10.5281/zenodo.23185896) Code and data:

<https://github.com/Ruqing1963/legendre-function-field>

Abstract

For a monic polynomial $f \in \mathbb{F}_q[t]$ of degree d , the *Legendre interval* $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$ is the function field analogue of $[n^2, (n+1)^2]$. We record the explicit form of a classical estimate (Hayes characters and Weil's theorem): $|\sum_{P \in \mathcal{I}_f} \Lambda(P) - q^{d+1}| \leq (d-2)(q^d - q)$ in every characteristic. Together with a bound for prime powers in $\mathcal{I}_f$, it shows that $\mathcal{I}_f$ contains an irreducible polynomial whenever $q \geq d - 1$. This supersedes all thresholds of earlier versions of this preprint, which overlooked it. The genuine analogue of Legendre's conjecture, with q fixed and $d \rightarrow \infty$, lies in the remaining range $q \leq d - 2$. We examine every Legendre interval for $q = 2$ with $d \leq 20$, for $q = 3$ with $d \leq 10$, for $q = 4$ with $d \leq 10$, and for $q = 5$ with $d \leq 8$. All of them contain irreducible polynomials, so the analogue of Legendre's conjecture holds for every q when $d \leq 8$. Over odd q the variance of the prime count across intervals is already close to the Keating–Rudnick prediction $(d-2)q^{d+1}$, proved only for $q \rightarrow \infty$, and the smallest normalized count approaches 1 as d grows. We state a conjecture with a heuristic based on these observations. We also give a self-contained proof that the geometric monodromy group of the family is S_{2d} for $p > 2d$, an exact twisted-variety identity, closed formulas for $d \leq 3$, and a list of corrections to v5 and v6.

Contents

1	Introduction	1
1.1	The question	1
1.2	What is known, and what this paper adds	2
1.3	About this revision	3
2	The classical range	3
2.1	Hayes classes	3
3	The open range $q \leq d - 2$	4
3.1	Method	4
3.2	Results	5
3.3	Observations	5
4	A conjecture	5

5	The monodromy group	6
5.1	Preliminaries	6
5.2	Full monodromy	7
6	The twisted root variety	8
7	Exact formulas for $d \leq 3$	8
8	Discussion	9
A	Errata to version v5	10
B	Corrections to version v6	10

1 Introduction

1.1 The question

Legendre's conjecture asserts that there is a prime between n^2 and $(n+1)^2$ for every $n \geq 1$. It is open. The best unconditional result on prime gaps gives primes in $[x, x + x^{0.525}]$ [1], and even the Riemann hypothesis only gives intervals of length about $\sqrt{x} \log x$.

In $\mathbb{F}_q[t]$ put $|g| = q^{\deg g}$. For monic f of degree d , the polynomials g with $|g - f^2| \leq |f|$ form the *Legendre interval*

$$\mathcal{I}_f = \{f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d\}, \quad |\mathcal{I}_f| = q^{d+1}.$$

Its elements are the monic polynomials of degree $2d$ whose coefficients of $t^{2d-1}, \dots, t^{d+1}$ agree with those of f^2 . We write

$$N_{\text{irr}}(f) = \#\{P \in \mathcal{I}_f : P \text{ irreducible}\}, \quad \Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P),$$

where $\Lambda(P) = \deg \pi$ if $P = \pi^r$ with π irreducible and $\Lambda(P) = 0$ otherwise. Then $2d N_{\text{irr}}(f) = \Psi(f) - E(f)$, where $E(f)$ collects the prime powers π^r , $r \geq 2$. The prime polynomial theorem predicts $N_{\text{irr}}(f) \approx q^{d+1}/(2d)$. *Does every Legendre interval contain an irreducible polynomial?*

1.2 What is known, and what this paper adds

The classical range $q \geq d - 1$. The interval $\mathcal{I}_f$ is a class of polynomials with $\ell = d - 1$ prescribed leading coefficients. Such classes are the fibres of Hayes's characters [9], whose L -functions are polynomials of degree at most $\ell - 1$ satisfying the Riemann hypothesis by Weil's theorem. This gives explicit bounds for irreducible polynomials with prescribed leading coefficients (Hsu [10], Cohen [4, Thm. 2.1]; see [7, Cors. 1–2]). For Legendre intervals they read as follows.

Theorem 1 (Section 2). *For every prime power q and every monic f of degree $d \geq 2$,*

$$|\Psi(f) - q^{d+1}| \leq (d-2)(q^d - q).$$

Consequently $N_{\text{irr}}(f) \geq 1$ whenever $q \geq d - 1$, in every characteristic.

The first statement is classical in substance; we give a short proof to fix constants. The consequence for $q \in \{d-1, d\}$ uses an elementary bound on $E(f)$ (Lemma 2.3). At $q = d-2$ the estimate gives nothing, because the main term q^{d+1} and the error $(d-2)q^d$ are then equal. This is the function field form of the familiar fact that the Riemann hypothesis *just* fails to reach Legendre’s conjecture over $\mathbb{Z}$.

Earlier versions of this preprint missed Theorem 1. Version v5 claimed the threshold $q > (8d)^{2d+6}$ without valid proof, and version v6 proved thresholds such as $q > 1.65 \times 10^4$ for $d = 4$. Both are far weaker than $q \geq d-1$; see Appendix B. The asymptotic $N_{\text{irr}} = q^{d+1}/2d + O_d(q^{d+1/2})$ for fixed d , the “main theorem” of v5, is contained in Theorem 1 and also in the short-interval theorem of Bank, Bary-Soroker and Rosenzweig [2]. Their result, and Sawin’s sharper bound [19], concern shorter intervals or larger q ; at the Legendre length the classical estimate is the relevant one.

The open range $q \leq d-2$. Here no general method is known. Reversing polynomials turns $\mathcal{I}_f$ into a residue class modulo t^d among polynomials of degree $2d$, at level of distribution exactly $1/2$. Sawin and Shusterman [20] pass level $1/2$ for squarefree moduli when q is large compared with p , but the moduli t^d are not covered. Results on prescribed coefficients [18, 8] allow far fewer than the $d-1$ coefficients needed here. We therefore study this range by exhaustive computation.

Theorem 2 (Section 3). *Every Legendre interval over $\mathbb{F}_q$ contains an irreducible polynomial when $q = 2, d \leq 20$; $q = 3, d \leq 10$; $q = 4, d \leq 10$; $q = 5, d \leq 8$. Together with Theorem 1, the function field analogue of Legendre’s conjecture holds for every prime power q when $d \leq 8$.*

The computations also measure how the prime count fluctuates from interval to interval. For odd q the Legendre intervals are *all* the intervals of length q^{d+1} in degree $2d$, and Keating and Rudnick [13] proved that the variance of Ψ over them is $\sim (d-2)q^{d+1}$ as $q \rightarrow \infty$. We find this already accurate for $q = 3$ and $q = 5$ (Table 1). A Gaussian model with this variance predicts that the smallest normalized count tends to 1 as $d \rightarrow \infty$, which is what the data show (Figure 1). This leads to:

Conjecture (Section 4). *For every prime power q and every $d \geq 1$, every Legendre interval in $\mathbb{F}_q[t]$ contains an irreducible polynomial. Moreover, for fixed q , $\min_f \Psi(f)/q^{d+1} \rightarrow 1$ as $d \rightarrow \infty$.*

Other results kept from v6. For $p > 2d$ we give a self-contained proof that the geometric monodromy group of $f^2 + \sum_{i \leq d} a_i t^i$ is S_{2d} , for every monic f (Theorem 5.7). We also prove an exact identity $\Psi(f) = \#W(\mathbb{F}_q)$ for an explicit twisted root variety W (Theorem 6.2), and closed formulas for N_{irr} when $d \leq 3$ (Section 7). These are not needed for Theorems 1 and 2, but they explain the factorization statistics in Figure 3 and give independent checks of the code.

1.3 About this revision

Appendix A lists the errors of v5, and Appendix B the omissions of v6 corrected here. The main changes from v6 are Theorem 1, which replaces all of v6’s explicit thresholds, and the new computations of Section 3.

2 The classical range

2.1 Hayes classes

For monic g of degree k put $\tilde{g}(u) = u^k g(1/u) \in \mathbb{F}_q[u]$; its constant term is 1. Let $\ell \geq 0$ and let $\mathcal{G}_\ell$ be the group of power series $1 + c_1 u + \dots$ modulo $u^{\ell+1}$, of order q^ℓ . The map

$g \mapsto [g] := \tilde{g} \bmod u^{\ell+1}$ is multiplicative, and $[g]$ records exactly the ℓ coefficients of g below the leading one. With $\ell = d - 1$, $\mathcal{I}_f$ is the set of monic P of degree $2d$ with $[P] = [f^2]$.

Lemma 2.1. *Let $\chi \neq 1$ be a character of $\mathcal{G}_\ell$ and $\chi(g) = \chi([g])$ for monic g . Then $L(z, \chi) = \sum_{g \text{ monic}} \chi(g) z^{\deg g}$ is a polynomial of degree at most $\ell - 1$, and its inverse roots have absolute value $q^{1/2}$ or 1.*

Proof. For $k \geq \ell$ the map $g \mapsto [g]$ from monic polynomials of degree k onto $\mathcal{G}_\ell$ has all fibres of size $q^{k-\ell}$, so $\sum_{\deg g=k} \chi(g) = 0$. Hence L is a polynomial of degree $\leq \ell - 1$. These are L -functions of characters of a ray class group of $\mathbb{F}_q(t)$, with conductor supported at the place $t = \infty$, and the Riemann hypothesis for them is Weil's theorem; see [9] and [13, §4], where they appear as even Dirichlet characters modulo $u^{\ell+1}$ after the substitution $t \mapsto 1/u$. $\square$

Theorem 2.2 (Theorem 1, first part). *For every q and every monic f of degree $d \geq 2$, $|\Psi(f) - q^{d+1}| \leq (d-2)(q^d - q)$. For $d = 2$ this means $\Psi(f) = q^3$ exactly.*

Proof. Let $\ell = d - 1$ and $n = 2d$. By orthogonality, $\Psi(f) = q^{-\ell} \sum_{\chi} \bar{\chi}([f^2]) \sum_{\deg P=n} \Lambda(P) \chi(P)$. The trivial character contributes $q^{-\ell} \cdot q^n = q^{d+1}$. For $\chi \neq 1$ write $L(z, \chi) = \prod_i (1 - \omega_i z)$ with at most $\ell - 1$ factors. Taking the logarithmic derivative gives $\sum_{\deg P=n} \Lambda(P) \chi(P) = -\sum_i \omega_i^n$, of absolute value at most $(\ell - 1)q^{n/2} = (d-2)q^d$. There are $q^\ell - 1$ nontrivial characters, each weighted by $q^{-\ell}$, so $|\Psi(f) - q^{d+1}| \leq (1 - q^{-(d-1)})(d-2)q^d = (d-2)(q^d - q)$. $\square$

Lemma 2.3 (Prime powers in a Legendre interval). *Let $E(f) = \Psi(f) - 2d N_{\text{irr}}(f)$. Then*

$$E(f) \leq dq^a + 2q^{\lfloor 2d/3 \rfloor}, \quad a = \begin{cases} 1, & p \text{ odd}, \\ \lceil (d+1)/2 \rceil, & p = 2. \end{cases}$$

Moreover $E(f) \leq \sum_{e|2d, e < 2d} q^e < q^d + 2q^{\lfloor 2d/3 \rfloor}$ in all cases.

Proof. $E(f)$ is the sum of $\deg \pi$ over irreducible π of degree $e = 2d/r$, $r \geq 2$, with $\pi^r \in \mathcal{I}_f$. Since $e \cdot \#\{\pi : \deg \pi = e\} \leq q^e$, the terms with $r \geq 3$ contribute at most $\sum_{e \leq 2d/3} q^e < 2q^{\lfloor 2d/3 \rfloor}$, and all terms together at most $\sum_{e|2d, e < 2d} q^e$. For $r = 2$, write $\pi = t^d + \pi_{d-1}t^{d-1} + \dots + \pi_0$. If p is odd, the coefficient of t^{2d-j} in π^2 is $2\pi_{d-j} + \Phi_j(\pi_{d-1}, \dots, \pi_{d-j+1})$ for a universal polynomial Φ_j , so the $d-1$ prescribed coefficients determine $\pi_{d-1}, \dots, \pi_1$, and at most q polynomials π qualify. If $p = 2$, then $\pi^2 = \sum_i \pi_i^2 t^{2i}$, the prescribed coefficients determine $\pi_{d-1}, \dots, \pi_{d-\lfloor (d-1)/2 \rfloor}$, and at most $q^{\lceil (d+1)/2 \rceil}$ polynomials qualify. Each contributes d . $\square$

Corollary 2.4 (Theorem 1, second part). *If $q \geq d - 1$, then $N_{\text{irr}}(f) \geq 1$ for every monic f of degree d .*

Proof. For $d \leq 2$ see Theorem 7.1. Let $d \geq 3$. By Theorem 2.2, $2d N_{\text{irr}}(f) \geq q^{d+1} - (d-2)(q^d - q) - E(f) = q^d(q - d + 2) + (d-2)q - E(f)$.

If $q \geq d$ the right side is at least $2q^d + (d-2)q - E(f)$, which is positive because $E(f) < q^d + 2q^{\lfloor 2d/3 \rfloor} \leq 2q^d$ (note $\lfloor 2d/3 \rfloor \leq d-1$ and $q \geq 2$).

If $q = d - 1$ the right side is $q^d + (d-2)q - E(f)$. For odd q (so $d \geq 4$), Lemma 2.3 gives $E(f) \leq dq + 2q^{d-2}$, since $\lfloor 2d/3 \rfloor \leq d-2$ for $d \geq 4$; and $2q + 2q^{d-2} < q^d$ because $q \geq 3$. For even q we have $d = q + 1$ odd and $E(f) \leq dq^{(d+1)/2} + 2q^{\lfloor 2d/3 \rfloor}$, which is less than q^d for $d \geq 5$, $q \geq 4$. The single remaining case $(q, d) = (2, 3)$ is checked directly: $N_{\text{irr}} = 2$ for every f (Table 1). $\square$

Remark 2.5. The second part of Theorem 1 is sharp for this method: when $q = d - 2$ the lower bound $q^d(q - d + 2) + (d-2)q - E(f)$ equals $q^2 - E(f)$, which is negative. Theorem 1 also gives $N_{\text{irr}} = q^{d+1}/2d + O(dq^d)$ for fixed d , with an explicit constant. This is stronger than the error $O_d(q^{d+1/2})$ of [2]. Sawin's bound [19] has the better q -exponent $q^{d/2+1}$ but a constant of size $(2d+2)^{3d}$; it is the right tool for q very large compared with d , not for existence.

All 174 exact counts with $q > d$ computed for v6 satisfy the bounds of Theorem 2.2 (data/interval_counts.csv).

3 The open range $q \leq d - 2$

3.1 Method

By Theorem 1 only the pairs with $q \leq d - 2$ need checking. The substitutions $t \mapsto \lambda t + b$ ($\lambda \in \mathbb{F}_q^\times$, $b \in \mathbb{F}_q$) preserve irreducibility and map Legendre intervals to Legendre intervals: f goes to $\lambda^{-d} f(\lambda t + b)$. We list all distinct Legendre intervals, group them into orbits under these substitutions, and for one representative of each orbit we test every element for irreducibility. We also compute $E(f)$ by enumerating the candidate prime powers, so $\Psi(f) = 2d N_{\text{irr}}(f) + E(f)$ is known exactly.

Field arithmetic uses addition and multiplication tables for $\mathbb{F}_q$, so prime and non-prime q are handled by the same code. Irreducibility is decided by Ben-Or's test: a monic P of degree n is irreducible if and only if $\gcd(t^{q^i} - t, P) = 1$ for $1 \leq i \leq n/2$. The implementation is cross-checked against an independent distinct-degree factorization for prime q , against a pure-Python implementation for $q = 4, 8, 9$, and against the twisted-variety count of Section 6. A further check is exact: for odd q every residue class of leading coefficients is a Legendre interval, so the orbit-weighted sum of Ψ must equal $q^{d-1} \cdot q^{d+1}$. It does in every case tested.

In characteristic 2 the Legendre intervals form a thin subfamily. Indeed $f^2 = \sum c_i^2 t^{2i}$, so every prescribed coefficient in an odd position is 0. There are only $q^{\lfloor (d-1)/2 \rfloor}$ Legendre intervals, which is why $q = 2$ can be pushed much further than $q = 3$.

3.2 Results

Theorem 3.1 (Theorem 2). *Every Legendre interval over $\mathbb{F}_q$ contains an irreducible polynomial for $q = 2$ and $d \leq 20$, $q = 3$ and $d \leq 10$, $q = 4$ and $d \leq 10$, and $q = 5$ and $d \leq 8$. Consequently, for every prime power q and every $d \leq 8$, every Legendre interval contains an irreducible polynomial.*

Proof. The first statement is the computation summarized in Table 1. For the second, let $d \leq 8$. If $q \geq d - 1$ apply Theorem 1. Otherwise $q \leq d - 2 \leq 8 - 2$, so q is one of the prime powers listed, and the first statement applies. $\square$

3.3 Observations

- (i) No empty Legendre interval occurs. The smallest normalized count in the open range is 0.375, at $(q, d) = (2, 6)$.
- (ii) For odd q the empirical variance of Ψ approaches $(d - 2)q^{d+1}$, although Keating and Rudnick prove the asymptotic only for $q \rightarrow \infty$. For $q = 3$ the relative deviation falls from 21% at $d = 6$ to 1.4% at $d = 10$; for $q = 5$ it is at most 11% for $5 \leq d \leq 8$.
- (iii) In characteristic 2 the Legendre intervals are a special subfamily and the variance fluctuates more. Small cases also show a downward bias in N_{irr} , for example $\min_f 2d N_{\text{irr}}/q^{d+1} = 0.375$ at $(q, d) = (2, 6)$. The bias fades as d grows.

4 A conjecture

Conjecture 4.1. *For every prime power q and every $d \geq 1$, every Legendre interval in $\mathbb{F}_q[t]$ contains an irreducible polynomial. Moreover $\min_f \Psi(f)/q^{d+1} \rightarrow 1$ as $d \rightarrow \infty$ with q fixed.*

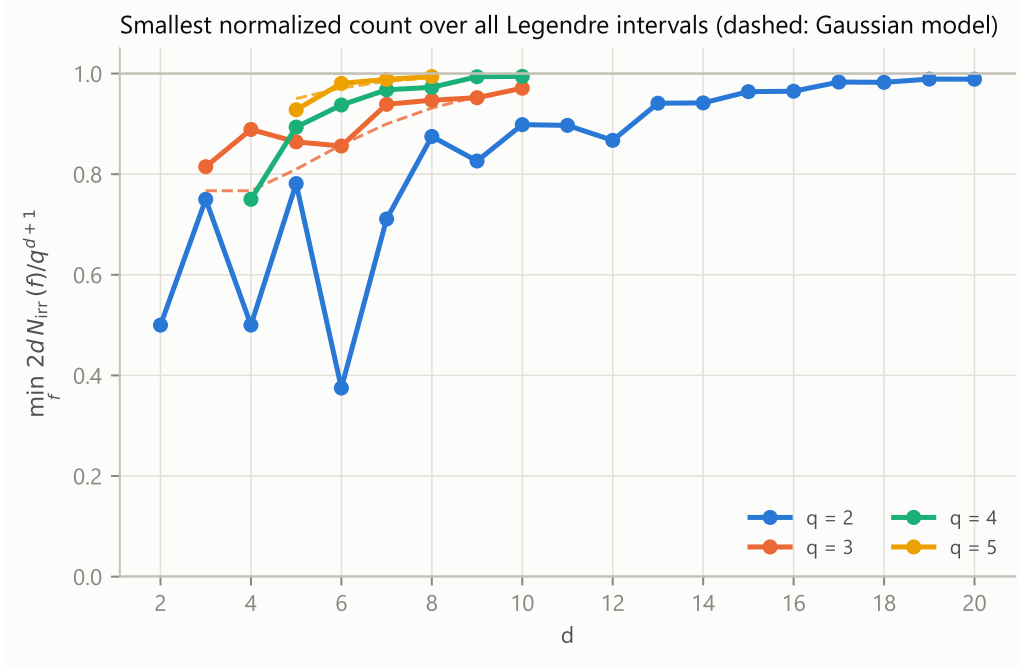


Figure 1: The smallest normalized count $\min_f 2d N_{\text{irr}}(f)/q^{d+1}$ in the open range. It never reaches 0 and approaches 1 as d grows.

Heuristic. Model the values $\Psi(f)$ over the q^{d-1} Legendre intervals (q odd) as Gaussian with mean q^{d+1} and variance $(d-2)q^{d+1}$, as suggested by [13] and by Table 1. The minimum of N such variables is about $-\sqrt{2 \log N}$ standard deviations, so

$$\min_f \frac{\Psi(f)}{q^{d+1}} \approx 1 - \sqrt{\frac{2(d-1)(d-2) \log q}{q^{d+1}}} \rightarrow 1.$$

The deficit is at most 0.51 over all $q \geq 2$ and $d \geq 3$, attained at $(q, d) = (2, 5)$, so the model predicts no empty interval at all. Since $E(f)$ is at most of order $q^{\lfloor 2d/3 \rfloor}$ for odd q , the same holds for N_{irr} . The observed minima are consistent with the prediction (Figure 1). The heuristic is not a proof. The Gaussian model is only known in the limit $q \rightarrow \infty$, and the extreme tail of the distribution is exactly what a proof would have to control.

5 The monodromy group

This section and the next two are kept from v6, with minor changes. Write $\mathcal{M} = \mathbb{A}^{d+1}$ with coordinates $\mathbf{a} = (a_0, \dots, a_d)$ and $F(t, \mathbf{a}) = f(t)^2 + \sum_{i \leq d} a_i t^i$.

5.1 Preliminaries

Lemma 5.1 (Membership in $\mathcal{I}_f$). *A monic $P \in \mathbb{F}_q[t]$ of degree $2d$ lies in $\mathcal{I}_f$ if and only if its coefficients of $t^{2d-1}, \dots, t^{d+1}$ agree with those of f^2 . If $p > d-1$, this holds if and only if the power sums $p_j(P) = \sum_{P(\rho)=0} \rho^j$ satisfy $p_j(P) = 2p_j(f)$ for $1 \leq j \leq d-1$.*

Proof. Newton's identities express the power sums $p_1, \dots, p_{d-1}$ through the top $d-1$ coefficients, and conversely when $1, \dots, d-1$ are invertible. Finally $p_j(f^2) = 2p_j(f)$. $\square$

Lemma 5.2 (Inertia at a simple node). *Let R be a complete discrete valuation ring with algebraically closed residue field of characteristic $\neq 2$, fraction field K and valuation v . Let $P \in R[t]$*

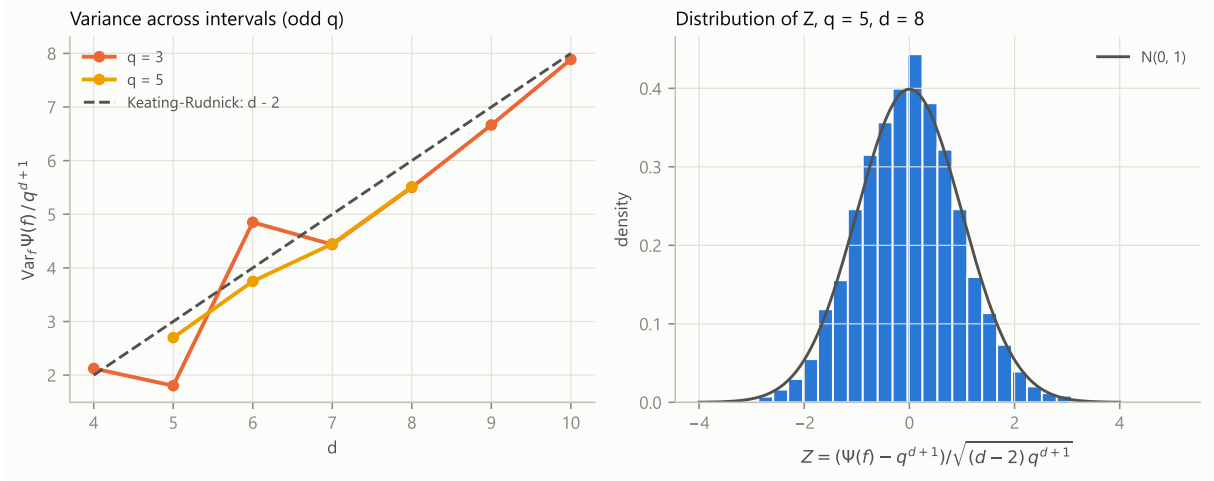


Figure 2: Left: $\text{Var}_f \Psi(f)/q^{d+1}$ over all Legendre intervals for odd q , against the Keating–Rudnick limit $d-2$. Right: the distribution of $Z = (\Psi(f) - q^{d+1})/\sqrt{(d-2)q^{d+1}}$ for the largest odd case, against the standard normal density.

be monic of degree n whose reduction $\bar{P}$ has exactly one double root β and $n-2$ simple roots, and suppose $v(\text{disc}P) = 1$. Then the splitting field of P over K is a quadratic extension whose Galois group acts on the roots of P as a transposition.

Proof. Write $\bar{P} = (t - \beta)^2 \bar{L}$ with $\bar{L}(\beta) \neq 0$ and $\bar{L}$ squarefree. By Hensel’s lemma $P = QL$ with $Q, L \in R[t]$ monic lifting $(t - \beta)^2$ and $\bar{L}$, and L splits into linear factors over R . Since $\text{disc}P = \text{disc}Q \cdot \text{disc}L \cdot \text{Res}(Q, L)^2$ and both $\text{disc}L$ and $\text{Res}(Q, L)$ are units, $v(\text{disc}Q) = 1$. An element of odd valuation is not a square in K , so Q is irreducible over K , and the nontrivial automorphism of its splitting field swaps the two roots of Q and fixes the roots of L . $\square$

Lemma 5.3 (Primitivity and decomposition; cf. [21, 17]). *Let K be a field, $G \in K[t]$ of degree $n \geq 2$ with $G' \neq 0$, and x transcendental over K . If $\text{Gal}(G(t) - x / K(x))$, acting on the roots of $G(t) - x$, is imprimitive, then $G = g \circ h$ with $g, h \in K[t]$ and $\deg g, \deg h \geq 2$.*

Proof. Let τ be a root. Blocks containing τ correspond to fields $K(x) \subseteq L \subseteq K(\tau)$, so imprimitivity gives a proper intermediate field L , and $L = K(h)$ by Lüroth’s theorem. Since $x = G(\tau)$ is a polynomial in τ , the place $\tau = \infty$ is the only place of $K(\tau)$ above $x = \infty$, and it is K -rational; so is its restriction to L . After a Möbius transformation over K we may take that place to be $h = \infty$. Then h has its only pole at $\tau = \infty$, so $h \in K[\tau]$, and likewise $x = g(h)$ with $g \in K[X]$. Both degrees are ≥ 2 because L is a proper intermediate field. $\square$

5.2 Full monodromy

Let $\mathcal{D} \subset \mathcal{M}$ be the zero locus of $\text{disc}_t F$ and $\mathcal{M}^\circ = \mathcal{M} \setminus \mathcal{D}$. The geometric monodromy group of the finite étale cover $\{F = 0\} \rightarrow \mathcal{M}^\circ$ is $G_{\text{geom}} = \text{Gal}(F/\mathbb{F}_q(\mathbf{a}))$. Put $K' = \mathbb{F}_q(a_1, \dots, a_d)$ and $G(t) = f(t)^2 + a_1 t + \dots + a_d t^d$, so $F = G(t) + a_0$ and $G_{\text{geom}} = \text{Gal}(G(t) - x/K'(x))$ with $x = -a_0$.

Proposition 5.4 (Transitivity). *F is irreducible over $\mathbb{F}_q(\mathbf{a})$, so G_{geom} is transitive.*

Proof. F has degree one in a_0 with coprime coefficients 1 and $f^2 + \sum_{i \geq 1} a_i t^i$, so it is irreducible in $\mathbb{F}_q[\mathbf{a}, t]$, hence in $\mathbb{F}_q(\mathbf{a})[t]$ by Gauss’s lemma. $\square$

Proposition 5.5 (Primitivity). *If $p > d$, then G_{geom} is primitive.*

Proof. For $d = 1$ this is automatic. Let $d \geq 2$; note $G' \neq 0$ because of the term $a_1 t$. If G_{geom} were imprimitive, Lemma 5.3 would give $G = g \circ h$ over K' with $\deg h = m$, $\deg g = n$, $mn = 2d$, $m, n \geq 2$, so $m \leq d$. Normalize h to be monic with $h(0) = 0$.

Step 1. For $1 \leq j \leq m - 1$ we have $2d - j \geq d + 1$, so the coefficient of t^{2d-j} in G is that of f^2 , an element of $\mathbb{F}_q$. Also $g(h) = h^n + (\text{terms of degree} \leq 2d - m)$ and $[t^{2d-j}]h^n = n h_{m-j} + \Phi_j(h_{m-1}, \dots, h_{m-j+1})$ with Φ_j universal. As $n \leq d < p$, induction gives $h \in \mathbb{F}_q[t]$.

Step 2. Let $U = \text{span}_{\mathbb{F}_q}\{1, h, \dots, h^n\}$. Then $G \in U \otimes K'$. For any linear form ℓ vanishing on U , $0 = \ell(f^2) + \sum_{i=1}^d a_i \ell(t^i)$, so $\ell(t) = 0$ because the a_i are algebraically independent. Hence $t \in U$, impossible since nonconstant elements of U have degree divisible by $m \geq 2$. $\square$

Proposition 5.6 (A transposition). *If $p > 2d$, then G_{geom} contains a transposition.*

Proof. Step 1: a fibre with exactly one double root. If $d = 1$ take $P^* = (t - \beta)^2$. For $d \geq 2$ fix $\beta \in \overline{\mathbb{F}}_q$, divide $f^2 = (t - \beta)^2 Q + r$ with $\deg r \leq 1$, and put $P^* = (t - \beta)^2(Q + c)$. Then $\deg(P^* - f^2) \leq 2 \leq d$, so P^* is a fibre of F . Since Q' has leading coefficient $2d - 2 \neq 0$ in $\mathbb{F}_q$, all but finitely many c make $R = Q + c$ squarefree with $R(\beta) \neq 0$.

Step 2: the discriminant vanishes to order one along $\varepsilon \mapsto \mathbf{a}^ + \varepsilon e_0$.* As $\partial_t F = P^{*'} is independent of ε with leading coefficient $2d \neq 0$, $\text{disc}_t(P^* + \varepsilon) = u \prod_{P^{*'}(\gamma)=0} (P^*(\gamma) + \varepsilon)$ with $u \neq 0$. Now $P^{*'} = (t - \beta)(2R + (t - \beta)R')$ has β as a simple root, and no root of R is a root of $P^{*'}$. So the order of vanishing at $\varepsilon = 0$ is exactly 1.$

Step 3. The inertia group at $\varepsilon = 0$ lies in G_{geom} and is generated by a transposition by Lemma 5.2. $\square$

Theorem 5.7. *If $p > 2d$, then $G_{\text{geom}} = S_{2d}$, and also $\text{Gal}(F/\mathbb{F}_q(\mathbf{a})) = S_{2d}$.*

Proof. A primitive permutation group containing a transposition is the full symmetric group [5, Thm. 3.3A]. The arithmetic group contains G_{geom} and lies in S_{2d} . $\square$

Remark 5.8. No squarefree hypothesis on f is needed. By [2, Prop. 3.6], $G_{\text{geom}} = S_{2d}$ also holds for every p when $d \geq 3$, and for odd p when $d = 2$.

6 The twisted root variety

Let e_j be the elementary symmetric polynomials in $u = (u_1, \dots, u_{2d})$ and $\gamma_j = (-1)^j [t^{2d-j}]f^2$. The root variety is $V = \{u \in \mathbb{A}^{2d} : e_j(u) = \gamma_j, 1 \leq j \leq d-1\}$. It is the variety $X_{n,m,\mathbf{c}}$ of [19] with $n = 2d$, $m = d-1$.

Proposition 6.1. *V is a reduced complete intersection of pure dimension $d+1$ and degree at most $(d-1)!$, and $V_{\overline{\mathbb{F}}_q}$ is irreducible if and only if $G_{\text{geom}} = S_{2d}$.*

Proof. Every component has dimension $\geq d+1$ by Krull's theorem and $\leq d+1$ because the map $u \mapsto \prod(t - u_i)$ is finite; the degree bound is the refined Bézout inequality [6, Ex. 8.4.6]. The map is étale where the u_i are distinct, so V is generically reduced, hence reduced. Over $\mathcal{M}^\circ$, V is an S_{2d} -torsor, and its number of geometric components is $[S_{2d} : G_{\text{geom}}]$. $\square$

Fix an $\mathbb{F}_q$ -basis $\omega_1, \dots, \omega_{2d}$ of $\mathbb{F}_{q^{2d}}$ with Moore matrix $M = (\omega_i^{q^{k-1}})_{k,i}$, and let $W \subset \mathbb{A}^{2d}$ be defined by $e_j(Mx) = \gamma_j$. Frobenius permutes the linear forms $(Mx)_k$ cyclically, so W is defined over $\mathbb{F}_q$; it is isomorphic to V over $\mathbb{F}_{q^{2d}}$, and $x \mapsto \sum x_i \omega_i$ identifies $W(\mathbb{F}_q)$ with $\{\beta \in \mathbb{F}_{q^{2d}} : \chi_\beta \in \mathcal{I}_f\}$, where $\chi_\beta = \prod_k (t - \beta^{q^k})$.

Theorem 6.2. *For every q and f : $\#W(\mathbb{F}_q) = \Psi(f)$, and $E(f)$ is the number of β in proper subfields of $\mathbb{F}_{q^{2d}}$ with $\chi_\beta \in \mathcal{I}_f$. Hence $2d N_{\text{irr}}(f) = \#W(\mathbb{F}_q) - E(f)$.*

Proof. If $\chi_\beta \in \mathcal{I}_f$ then $\chi_\beta = \pi^r$ with π the minimal polynomial of β , and each π arises from exactly $\deg \pi$ elements β . Irreducible $P \in \mathcal{I}_f$ correspond to $r = 1$. $\square$

Remark 6.3. Theorem 6.2 turns $\Psi(f)$ into a point count on an explicit variety. With the Lang–Weil bound of Cafure and Matera [3] it gave the thresholds of v6, which Theorem 1 now supersedes (Appendix B). The identity remains useful for exact formulas and as an independent test of the code.

7 Exact formulas for $d \leq 3$

Theorem 7.1. (i) If $d = 1$ then $N_{\text{irr}} = (q^2 - q)/2$.

(ii) If $d = 2$ then $N_{\text{irr}} = (q^3 - q)/4$ for odd q and $N_{\text{irr}} = (q^3 - q^2)/4$ for even q .

(iii) If $d = 3$, $p \geq 5$, $f = t^3 + c_2 t^2 + c_1 t + c_0$, $\Delta = c_2^2 - 3c_1$ and η is the quadratic character of $\mathbb{F}_q$, then

$$6 N_{\text{irr}} = \begin{cases} q^4 - \eta(2\Delta) q^2 - q + \eta(-3) + \eta(2\Delta) - 1, & \Delta \neq 0, \\ q^4 - q - (q - 1) \eta(-3), & \Delta = 0. \end{cases}$$

Proof. (i) $\mathcal{I}_f$ is the set of all monic quadratics. (ii) By Theorem 2.2, $\Psi(f) = q^3$. The prime powers in $\mathcal{I}_f$ are the squares π^2 with $\deg \pi = 2$ or $\pi = (t - b)$; by Theorem 6.2, $E(f)$ is the number of $\beta \in \mathbb{F}_{q^2}$ with $\chi_\beta \in \mathcal{I}_f$. For such β , $\chi_\beta = (t^2 - \text{Tr} \beta t + N\beta)^2$ has t^3 -coefficient $-2\text{Tr} \beta$, which must equal $[t^3]f^2 = 2c_1$. For odd q this has q solutions; for even q both sides vanish and all q^2 elements qualify. (iii) This is the quadric count of v6, Theorem 6.2: W is a quadric in a hyperplane of $\mathbb{A}^6$, and the counts for nondegenerate quadratic forms [15, Thms. 6.26, 6.27] give $\#W(\mathbb{F}_q)$ and $E(f)$ exactly. It agrees with Kuz'min's formula for two prescribed coefficients [14, 16]. $\square$

All 141 applicable exact counts of `data/interval_counts.csv` agree with (i)–(iii), and (ii) for even q was checked for $q = 2, 4, 8$.

8 Discussion

Where the difficulty lies. Theorem 1 shows that the analogue of Legendre's conjecture is a consequence of the Riemann hypothesis for Hayes L -functions as long as $q \geq d - 1$. Below that, the $d - 2$ inverse roots of each of the $q^{d-1} - 1$ L -functions could in principle conspire, and nothing rules this out. Over the integers the corresponding statement is that the Riemann hypothesis gives primes in $[x, x + c\sqrt{x} \log x]$ but not in $[x, x + 2\sqrt{x}]$. In both settings the missing input is cancellation among zeros, not the location of zeros.

What the data suggest. The variance across intervals matches the random-matrix prediction of Keating and Rudnick even for $q = 3$, so the zeros of the relevant L -functions do behave independently in the range we can compute. If that persists, empty intervals never occur (Section 4).

Open problems.

- (1) Prove Conjecture 4.1 for one fixed q , for instance $q = 2$, where the Legendre intervals form a thin and structured family.
- (2) Prove the Keating–Rudnick variance for fixed q and $d \rightarrow \infty$, even in the Legendre range.
- (3) Extend Theorem 2 to $d = 9$. This requires $q = 5$ and $q = 7$ with $d = 9$ in full, about 2×10^{11} and 4×10^{13} irreducibility tests with the present method.

Factorization types in $\mathcal{I}_{t^d}$ follow the cycle types of S_{2d}

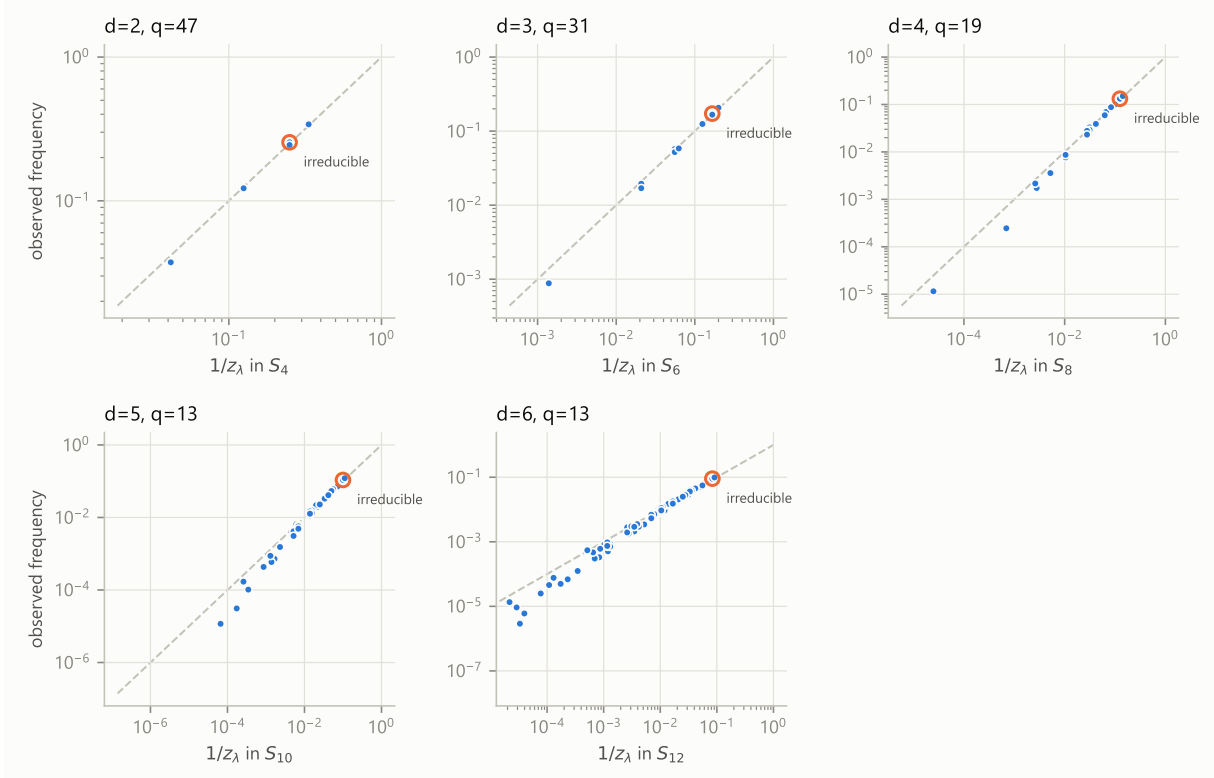


Figure 3: Observed frequency of each factorization type in $\mathcal{I}_{t^d}$ against the S_{2d} cycle-type probability $1/z_\lambda$, as predicted by Theorem 5.7 and Chebotarev. The ringed point is the irreducible type.

- (4) Higher powers: the intervals $\{f^k + s : \deg s \leq (k-1)d\}$ for $k \geq 3$, where the classical estimate covers a different range.

A Errata to version v5

- E1. **Novelty.** The main theorem of v5, $N_{\text{irr}} = q^{d+1}/(2d) + O_d(q^{d+1/2})$, follows from the classical estimate of Theorem 1 and from [2, Thm. 2.3], without the squarefree hypothesis and in every odd characteristic.
- E2. **Primitivity.** v5 applied “imprimitive $\Leftrightarrow F = g \circ h$ ” to the Galois group of F over $\overline{\mathbb{F}}_q(\mathbf{a})$, where it does not hold as stated, and treated algebraic coefficients as free parameters. Fixed in Proposition 5.5.
- E3. **Transposition.** v5 misidentified the discriminant branches near $\mathbf{a} = 0$ and invoked Picard–Lefschetz without justification. Fixed in Proposition 5.6.
- E4. **Effective Chebotarev.** v5 stated the bound with the permutation sheaf, which detects linear factors, not $2d$ -cycles, and cited an inapplicable theorem of [11].
- E5. **Betti-number bound.** v5 attributed to [12, Thm. 12] a bound for push-forwards of finite covers that is not there. The threshold $(8d)^{2d+6}$ was unproved.
- E6. **Other.** An error-term mix-up in the final step, a j_1/j_* slip in the Grothendieck–Ogg–Shafarevich formula, wrong data for the Bary-Soroker citation, uncited references, version-

history remarks in the body, and conclusions drawn from three small examples.

B Corrections to version v6

- F1. **Classical estimate omitted.** v6 did not mention the Hayes–Weil estimate (Theorem 1, [4, 10, 7]). It gives $N_{\text{irr}} \geq 1$ for $q \geq d - 1$ and error $O(dq^d)$. v6’s Theorem C ($q > 1.65 \times 10^4$ for $d = 4$), its Katz–Deligne bound and its bound via [19] (threshold about $(2d + 2)^6$) are all superseded for existence; Table 2 compares them.
- F2. **Level 1/2.** v6 said that at level 1/2 the Riemann hypothesis gives an error larger than the main term “by a factor of order d ”. The factor is d/q , as Theorem 1 shows.
- F3. **Novelty of the twisted identity.** The root variety is Sawin’s $X_{n,m,c}$, so Theorem 6.2 is a twisted form of his point count (this was added late in v6).
- F4. **Scope.** v6 described q fixed, $d \rightarrow \infty$ as open without locating the open range; it is $q \leq d - 2$.

References

- [1] R. C. Baker, G. Harman, J. Pintz, The difference between consecutive primes, II, *Proc. London Math. Soc.* **83** (2001), 532–562.
- [2] E. Bank, L. Bary-Soroker, L. Rosenzweig, Prime polynomials in short intervals and in arithmetic progressions, *Duke Math. J.* **164** (2015), 277–295. arXiv:1302.0625.
- [3] A. Cafure, G. Matera, Improved explicit estimates on the number of solutions of equations over a finite field, *Finite Fields Appl.* **12** (2006), 155–185.
- [4] S. D. Cohen, Explicit theorems on generator polynomials, *Finite Fields Appl.* **11** (2005), 337–357.
- [5] J. D. Dixon, B. Mortimer, *Permutation Groups*, GTM 163, Springer, 1996.
- [6] W. Fulton, *Intersection Theory*, 2nd ed., Springer, 1998.
- [7] Z. Gao, Improved error bounds for the number of irreducible polynomials and self-reciprocal irreducible monic polynomials with prescribed coefficients over a finite field, arXiv:2109.14154, 2021.
- [8] J. Ha, Irreducible polynomials with several prescribed coefficients, *Finite Fields Appl.* **40** (2016), 10–25.
- [9] D. R. Hayes, The distribution of irreducibles in $GF[q, x]$, *Trans. Amer. Math. Soc.* **117** (1965), 101–127.
- [10] C. N. Hsu, The distribution of irreducible polynomials in $\mathbb{F}_q[t]$, *J. Number Theory* **61** (1996), 85–96.
- [11] N. M. Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Ann. of Math. Stud. 116, Princeton Univ. Press, 1988.
- [12] N. M. Katz, Sums of Betti numbers in arbitrary characteristic, *Finite Fields Appl.* **7** (2001), 29–44.

- [13] J. P. Keating, Z. Rudnick, The variance of the number of prime polynomials in short intervals and in residue classes, *Int. Math. Res. Not.* 2014, 259–288. arXiv:1204.0708.
- [14] M. Lalin, O. Larocque, The number of irreducible polynomials with the first two prescribed coefficients over a finite field, *Rocky Mountain J. Math.* **46** (2016), 1587–1618.
- [15] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge Univ. Press, 1997.
- [16] G. McGuire, E. S. Yilmaz, The number of irreducible polynomials with the first two coefficients fixed over finite fields of odd characteristic, arXiv:1609.02314.
- [17] P. Müller, Primitive monodromy groups of polynomials, in *Recent Developments in the Inverse Galois Problem*, Contemp. Math. **186**, Amer. Math. Soc., 1995, 385–401.
- [18] P. Pollack, Irreducible polynomials with several prescribed coefficients, *Finite Fields Appl.* **22** (2013), 70–78.
- [19] W. Sawin, Square-root cancellation for sums of factorization functions over short intervals in function fields, *Duke Math. J.* **170** (2021), 997–1026. arXiv:1809.05137.
- [20] W. Sawin, M. Shusterman, On the Chowla and twin primes conjectures over $\mathbb{F}_q[T]$, *Ann. of Math.* **196** (2022), no. 2. arXiv:1808.04001.
- [21] G. Turnwald, On Schur’s conjecture, *J. Austral. Math. Soc. Ser. A* **58** (1995), 312–357.

q	d	intervals	orbits	$\min N_{\text{irr}}$	ratio	$\text{Var } \Psi/q^{d+1}$	$d - 2$
2	2	1	1	1	0.500	0.00	–
2	3	2	1	2	0.750	0.00	–
2	4	2	2	2	0.500	2.00	2
2	5	4	2	5	0.781	1.25	3
2	6	4	3	4	0.375	15.56	4
2	7	8	4	13	0.711	7.72	5
2	8	8	6	28	0.875	1.59	6
2	9	16	8	47	0.826	5.77	7
2	10	16	10	92	0.898	7.91	8
2	11	32	16	167	0.897	15.35	9
2	12	32	20	296	0.867	20.74	10
2	13	64	32	593	0.941	11.47	11
2	14	64	36	1102	0.942	17.37	12
2	15	128	64	2106	0.964	11.86	13
2	16	128	72	3952	0.965	15.22	14
2	17	256	128	7580	0.983	13.88	15
2	18	256	136	14310	0.983	19.71	16
2	19	512	256	27290	0.989	17.65	17
2	20	512	272	51841	0.989	19.45	18
3	5	81	18	63	0.864	1.80	3
3	6	243	48	156	0.856	4.85	4
3	7	729	135	440	0.939	4.44	5
3	8	2187	378	1165	0.947	5.50	6
3	9	6561	1143	3123	0.952	6.66	7
3	10	19683	3321	8600	0.971	7.89	8
3	11 [†]	59049	30	23996	0.993	–	9
4	6	16	3	1280	0.938	9.01	4
4	7	64	8	4530	0.968	9.36	5
4	8	64	12	15936	0.973	13.25	6
4	9	256	24	57886	0.994	6.36	7
4	10	256	28	208536	0.994	10.47	8
5	7	15625	815	27579	0.988	4.45	5
5	8	78125	3940	121334	0.994	5.51	6

Table 1: The open range. “Intervals” is the number of distinct Legendre intervals and “orbits” the number examined after the affine reduction; rows marked [†] examine a random sample of orbits. The ratio is $\min_f 2d N_{\text{irr}}(f)/q^{d+1}$. The variance column is the mean of $(\Psi(f) - q^{d+1})^2/q^{d+1}$ over all Legendre intervals. For odd q it is to be compared with the Keating–Rudnick limit $d - 2$ [13], valid for $d \geq 4$; for $q = 2, 4$ the Legendre intervals are a special subfamily and no prediction applies. Rows with $q \geq d - 1$ other than $q = 2$ are omitted, since Theorem 1 covers them. Source: `data/open_regime_summary.csv`.

d	classical	v5 (unproved)	Katz + Deligne	Cafure–Matera	via Sawin
1	–	7.2	2.2	0.8	–
2	–	12.0	5.7	0.9	4.7
3	0.00	16.6	10.6	2.0	5.3
4	0.30	21.1	16.3	4.2	5.9
5	0.48	25.6	22.7	6.9	6.4
6	0.60	30.3	29.7	9.9	6.8
8	0.78	39.7	44.8	16.8	7.4
10	0.90	49.5	61.4	24.8	7.9
15	1.11	74.9	106.9	48.1	8.9
20	1.26	101.4	157.0	74.7	9.7
30	1.45	157.1	266.4	134.8	10.7
40	1.58	215.4	384.8	201.4	11.4
50	1.68	275.8	509.8	272.8	12.0

Classical: $N_{\text{irr}} \geq 1$ for $q > d - 2$ (Theorem 1); for $d \leq 2$ for every q .

Table 2: Thresholds $\log_{10} q_0(d)$ beyond which $N_{\text{irr}} \geq 1$ is proved. “Classical” is Theorem 1 ($q_0 = d - 2$). The other rigorous columns are the bounds of v6: Katz’s Betti bound with Deligne, Cafure–Matera, and Sawin’s theorem. “v5” was never proved.